

UNIT I

BASICS OF COMPUTER NETWORK

Topics Covered

- 1.1 Computer Networking
 - 1.2 Signals — Analog and Digital Signals
 - 1.3 Parallel and Serial Transmission Mode
 - 1.4 Data Transmission Media
 - 1.5 Network Topologies – Bus, Star, Ring, Mesh
 - 1.6 Network Types – LAN, MAN, WAN
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1.1 COMPUTER NETWORKING

What is a Computer Network?

A **computer network** is a collection of two or more computers and other devices connected together to communicate with each other and share resources.

A network allows computers to exchange:

- Data
- Files
- Messages
- Hardware resources
- Software resources
- Internet connections

Examples of network devices include:

- Computers
- Laptops
- Smartphones
- Printers
- Servers
- Routers
- Switches

Simple Example

When several computers in a college laboratory are connected together, they form a **computer network**. The computers can share files, access a common printer, and communicate with a server.

Definition of Computer Networking

Computer networking is the process of connecting computers and other devices so that they can communicate and share data and resources with each other.

Components of a Computer Network

A computer network generally consists of the following components:

1. Sender

The sender is the device that sends or transmits data.

Example:

- Computer
- Mobile phone
- Server

2. Receiver

The receiver is the device that receives the transmitted data.

Example:

- Computer
- Printer
- Mobile phone

3. Transmission Medium

The transmission medium is the path through which data travels from sender to receiver.

Examples:

- Twisted-pair cable
- Coaxial cable
- Optical fiber
- Radio waves

4. Network Devices

Network devices help connect and manage communication between computers.

Examples:

- Hub
- Switch
- Router

- Modem
- Access Point

5. Protocols

Protocols are a set of rules that define how devices communicate with each other.

Examples:

- TCP/IP
 - HTTP
 - FTP
 - SMTP
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Characteristics of Computer Networks

Important characteristics of computer networks include:

1. Resource Sharing

Networks allow users to share resources such as:

- Printers
- Storage
- Internet connections
- Software

2. Data Sharing

Users can easily exchange files and information over a network.

3. Communication

Users can communicate using:

- Email
- Chat
- Video conferencing
- Voice calls

4. Centralized Management

Network administrators can manage users, devices, and resources from a central location.

5. Reliability

Data can sometimes be accessed through alternative network paths if one path fails.

6. Scalability

Networks can be expanded by adding more computers and devices.

Advantages of Computer Networking

1. **Resource Sharing** – Hardware and software resources can be shared.
 2. **File Sharing** – Users can easily exchange files.
 3. **Communication** – Enables fast communication between users.
 4. **Cost Reduction** – Shared resources reduce expenses.
 5. **Centralized Data Management** – Data can be stored and managed centrally.
 6. **Internet Sharing** – Multiple devices can use a common Internet connection.
 7. **Remote Access** – Users can access resources from different locations.
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Disadvantages of Computer Networking

1. **Security Risks** – Unauthorized users may access network resources.
 2. **Virus and Malware Spread** – Malware can spread quickly through connected devices.
 3. **Network Failure** – A network failure may affect multiple users.
 4. **Maintenance Cost** – Networks require maintenance and administration.
 5. **Privacy Issues** – Data transmitted over a network may be exposed if not properly secured.
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1.2 SIGNALS – ANALOG AND DIGITAL SIGNALS

What is a Signal?

A **signal** is an electrical or electromagnetic representation used to carry information from one location to another.

In data communication, signals are used to transmit data through a communication medium.

Signals are mainly classified into:

1. Analog Signal
 2. Digital Signal
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A. ANALOG SIGNAL

Definition

An **analog signal** is a continuous signal that changes smoothly over time.

It can have an infinite number of possible values within a given range.

Analog signals are commonly represented as continuous waves.

Examples

- Human voice
 - Traditional telephone signals
 - Radio signals
 - Analog television signals
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Characteristics of Analog Signals

1. Analog signals are continuous.
 2. They can have infinite values.
 3. They are represented by continuous waveforms.
 4. They are more affected by noise and interference.
 5. Signal quality may decrease over long distances.
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Components of an Analog Signal

The main characteristics of an analog signal are:

1. Amplitude

Amplitude represents the strength or maximum height of the signal.

2. Frequency

Frequency represents the number of cycles completed by a signal in one second.

It is measured in **Hertz (Hz)**.

3. Phase

Phase represents the position of a waveform relative to a reference point.

B. DIGITAL SIGNAL

Definition

A **digital signal** represents data using discrete values.

In most computer systems, digital signals use two levels:

- 0
- 1

These represent binary data.

Digital signals are commonly represented using square or rectangular waveforms.

Examples

- Computer data
- Digital audio
- Digital video
- Data transmitted through computer networks

Characteristics of Digital Signals

1. Digital signals are discrete.
2. They commonly use binary values.
3. They are less affected by noise compared with analog signals.
4. They are easier to store and process using computers.
5. Digital signals can be regenerated accurately.

Difference Between Analog and Digital Signals

Analog Signal	Digital Signal
Continuous signal.	Discrete signal.
Has infinite possible values.	Has a limited number of discrete values.
Usually represented by smooth waves.	Usually represented by pulses or square waves.
More affected by noise.	Generally more resistant to noise.
Difficult to process directly using digital computers.	Easy to process using digital computers.
Example: Traditional voice signal.	Example: Computer binary data.

1.3 PARALLEL AND SERIAL TRANSMISSION MODE

Data transmission refers to the process of transferring data from one device to another.

Based on how bits are transmitted, data transmission can be classified as:

1. Parallel Transmission
2. Serial Transmission

A. PARALLEL TRANSMISSION

Definition

In **parallel transmission**, multiple bits are transmitted simultaneously using multiple communication lines.

For example, if 8 bits are transmitted in parallel, 8 bits can be sent at the same time using separate lines.

Example

Suppose the data is:

10110010

All 8 bits can be transmitted simultaneously over 8 separate lines.

Advantages of Parallel Transmission

1. Multiple bits are transmitted at the same time.
 2. It can provide high data transfer speed over short distances.
 3. It is suitable for communication within a computer system.
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Disadvantages of Parallel Transmission

1. Requires multiple communication lines.
 2. More expensive for long-distance communication.
 3. Signals may arrive at slightly different times, causing timing problems.
 4. Not generally suitable for long-distance communication.
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Applications of Parallel Transmission

Parallel transmission has historically been used in:

- Internal computer buses
 - Older printer interfaces
 - Short-distance data communication
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B. SERIAL TRANSMISSION

Definition

In **serial transmission**, bits are transmitted one after another over a communication channel.

For example, the binary data:

10110010

is transmitted bit by bit in sequence.

Advantages of Serial Transmission

1. Requires fewer communication lines.
 2. Less expensive for long-distance communication.
 3. Suitable for communication over long distances.
 4. Easier to implement over modern communication networks.
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Disadvantages of Serial Transmission

1. Bits are transmitted sequentially.
 2. It may require additional mechanisms for synchronization.
 3. Depending on the technology, transmission may be slower than parallel transmission over very short distances.
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Applications of Serial Transmission

Examples include:

- USB
 - Ethernet
 - Serial communication interfaces
 - Long-distance communication systems
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Difference Between Parallel and Serial Transmission

Parallel Transmission	Serial Transmission
Multiple bits are transmitted simultaneously.	Bits are transmitted one after another.
Uses multiple communication lines.	Usually uses fewer communication lines.
Suitable for short distances.	Suitable for longer distances.
More expensive for long-distance communication.	Generally more economical.
Can experience timing/skew problems between lines.	Avoids parallel line skew.
Example: Computer internal buses.	Example: USB and Ethernet communication.

1.4 DATA TRANSMISSION MEDIA

What is Transmission Media?

Transmission media refers to the physical or wireless path through which data signals travel from a sender to a receiver.

Transmission media are mainly divided into:

1. Guided Media
2. Unguided Media

A. GUIDED TRANSMISSION MEDIA

Guided media uses a physical path or cable to transmit signals.

The main types are:

1. Twisted Pair Cable
2. Coaxial Cable
3. Optical Fiber Cable

1. Twisted Pair Cable

A twisted-pair cable consists of two insulated copper wires twisted together.

The twisting helps reduce electromagnetic interference.

There are two common types:

a. UTP – Unshielded Twisted Pair

UTP does not have additional metallic shielding.

Advantages:

- Low cost
- Easy to install
- Flexible
- Commonly used in Ethernet networks

Disadvantages:

- More susceptible to interference than shielded cables
- Limited distance compared with fiber

b. STP – Shielded Twisted Pair

STP contains additional shielding to reduce electromagnetic interference.

Advantages:

- Better protection against interference
- Suitable for environments with electromagnetic noise

Disadvantages:

- More expensive
 - Less flexible than UTP
 - More difficult to install
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2. Coaxial Cable

A coaxial cable consists of:

- Central conductor
- Insulating layer
- Metallic shield
- Outer protective covering

Coaxial cables provide better shielding against interference than basic twisted-pair cables.

Applications

- Cable television
 - Broadband Internet
 - Some communication systems
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3. Optical Fiber Cable

Optical fiber uses **light signals** to transmit data through thin strands of glass or plastic.

It provides very high bandwidth and supports long-distance communication.

Advantages

1. Very high bandwidth.
2. High data transmission speed.
3. Less affected by electromagnetic interference.
4. Suitable for long-distance communication.
5. More difficult to tap physically than copper cables.

Disadvantages

1. More expensive to install.
2. Requires specialized equipment.
3. Fiber can be more delicate to handle than copper cable.

Applications

- Internet backbone networks
 - Telephone networks
 - Cable television
 - Data centers
 - Long-distance communication
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B. UNGUIDED TRANSMISSION MEDIA

Unguided media transmits signals through air or space without using a physical cable.

The main types include:

1. Radio Waves
 2. Microwaves
 3. Infrared
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1. Radio Waves

Radio waves are electromagnetic waves used for wireless communication.

They can travel through air and, depending on frequency and conditions, can cover large distances.

Applications

- Radio broadcasting
 - Wireless networks
 - Mobile communication
 - IoT devices
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2. Microwaves

Microwaves are high-frequency electromagnetic waves commonly used for communication over line-of-sight paths.

Applications

- Satellite communication

- Mobile networks
 - Point-to-point communication
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3. Infrared

Infrared communication uses infrared light for short-range wireless communication.

It generally requires the communicating devices to be relatively close and may be blocked by obstacles.

Applications

- Remote controls
 - Short-range device communication
 - Some sensors
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Difference Between Guided and Unguided Media

Guided Media	Unguided Media
Uses a physical transmission path.	Uses air or space.
Includes cables and wires.	Uses wireless signals.
Generally more secure physically.	Can be more exposed to interception.
Installation may require physical cabling.	Does not require physical cables between endpoints.
Examples: Twisted pair, coaxial, fiber.	Examples: Radio, microwave, infrared.

1.5 NETWORK TOPOLOGIES

What is Network Topology?

Network topology refers to the physical or logical arrangement of devices and communication links in a computer network.

It describes how computers, network devices, and communication paths are connected.

Common network topologies include:

1. Bus Topology
 2. Star Topology
 3. Ring Topology
 4. Mesh Topology
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A. BUS TOPOLOGY

Definition

In a **Bus Topology**, all devices are connected to a single main communication cable called the **backbone**.

Data transmitted by one device travels along the shared cable, and the intended device receives it.

Simple Representation

Computer A

|

|

=====

Main Bus Cable

=====

| |

Computer B Computer C

Advantages of Bus Topology

1. Simple to install.
2. Requires less cable.
3. Relatively inexpensive for small networks.
4. Easy to understand.

Disadvantages of Bus Topology

1. Failure of the main cable can affect the entire network.
2. Troubleshooting can be difficult.
3. Network performance may decrease as more devices are added.
4. Data collisions may occur in shared-medium networks.

Applications

Bus topology was commonly used in older Ethernet networks.

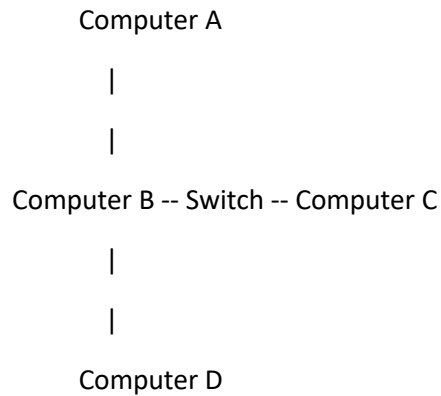
B. STAR TOPOLOGY

Definition

In a **Star Topology**, all devices are connected to a central device such as a switch or hub.

The central device manages communication between connected devices.

Simple Representation



Advantages of Star Topology

1. Easy to install and manage.
 2. Easy to identify individual connection failures.
 3. Failure of one cable usually affects only one device.
 4. Easy to add or remove devices.
 5. Commonly used in modern Ethernet LANs.
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Disadvantages of Star Topology

1. Requires more cable than bus topology.
 2. Requires a central device.
 3. Failure of the central device can affect the entire network.
 4. Installation cost can be higher than simple bus networks.
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C. RING TOPOLOGY

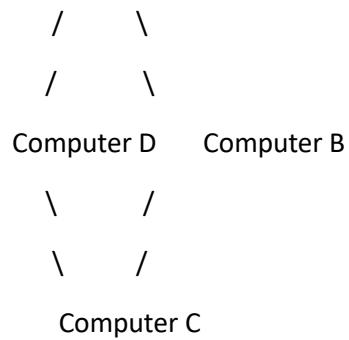
Definition

In a **Ring Topology**, each device is connected to two neighboring devices, forming a circular path.

Data travels around the ring according to the communication method used by the network.

Simple Representation

Computer A



Advantages of Ring Topology

1. Data transmission can be organized in a predictable manner.
 2. Each device has a defined connection to neighboring devices.
 3. Can provide orderly access to the communication medium.
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Disadvantages of Ring Topology

1. Failure of a connection can affect communication around the ring.
 2. Adding or removing devices may interrupt the network.
 3. Troubleshooting can be difficult.
 4. Less common in modern local area networks.
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D. MESH TOPOLOGY

Definition

In a **Mesh Topology**, devices are connected through multiple communication links.

In a **Full Mesh**, every device has a direct connection to every other device.

In a **Partial Mesh**, only some devices have multiple direct connections.

Simple Representation

Computer A ----- Computer B



Computer C ----- Computer D

Advantages of Mesh Topology

1. Provides high redundancy.
2. Multiple communication paths are available.
3. Failure of one link may not stop all communication.
4. Provides high reliability.
5. Suitable for critical network infrastructure.

Disadvantages of Mesh Topology

1. Requires a large amount of cabling in full mesh networks.
2. Installation is complex.
3. Maintenance can be difficult.
4. More expensive than simpler topologies.

Comparison of Network Topologies

Feature	Bus	Star	Ring	Mesh
Structure	Single backbone cable	Central device	Circular connection	Multiple interconnections
Cost	Low	Medium	Medium	High
Installation	Easy	Easy	Moderate	Difficult
Reliability	Low to Medium	Medium to High	Medium	Very High
Main Failure Point	Backbone cable	Central device	Link/device failure can affect ring	Individual link failure may be tolerated
Scalability	Limited	Good	Moderate	Complex
Common Use	Older networks	Modern LANs	Specialized/older networks	Critical networks

1.6 NETWORK TYPES: LAN, MAN, WAN

Computer networks can be classified according to the geographical area they cover.

The major types are:

1. LAN – Local Area Network
 2. MAN – Metropolitan Area Network
 3. WAN – Wide Area Network
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A. LAN – LOCAL AREA NETWORK

Definition

A **Local Area Network (LAN)** is a computer network that covers a small geographical area.

Examples include:

- Home
- Office
- School
- College
- Computer laboratory
- Small building

A LAN is generally owned and managed by a single organization or individual.

Example

A college computer laboratory where all computers are connected to a common network is an example of a LAN.

Characteristics of LAN

1. Covers a small geographical area.
 2. Usually provides high data transfer rates.
 3. Generally has low latency.
 4. Usually privately owned.
 5. Easy to manage and maintain.
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Advantages of LAN

- Fast communication
- Resource sharing
- File sharing
- Printer sharing

- Internet connection sharing
 - Centralized management
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B. MAN – METROPOLITAN AREA NETWORK

Definition

A **Metropolitan Area Network (MAN)** is a network that covers a larger geographical area than a LAN, usually across a city or metropolitan region.

A MAN can connect multiple LANs within a city.

Example

A network connecting branches of an organization located in different parts of the same city can be considered a MAN.

Characteristics of MAN

1. Covers a city or metropolitan area.
 2. Larger than a LAN.
 3. Smaller than a WAN.
 4. Can connect multiple LANs.
 5. May be operated by organizations or service providers.
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C. WAN – WIDE AREA NETWORK

Definition

A **Wide Area Network (WAN)** is a network that covers a large geographical area.

It may connect:

- Cities
- States
- Countries
- Continents

The **Internet** is the best-known example of a global network based on interconnected networks.

Characteristics of WAN

1. Covers a very large geographical area.
2. Can connect multiple LANs and MANs.

3. Uses various communication technologies.
4. May use leased lines, fiber, microwave, satellite, or other communication links.
5. Usually involves multiple organizations or service providers.

Advantages of WAN

1. Connects geographically distant locations.
2. Enables communication between branches.
3. Allows centralized access to resources.
4. Supports remote access.
5. Facilitates global communication.

Disadvantages of WAN

1. More expensive to establish and maintain.
2. More complex to manage.
3. Security risks are greater due to wider geographical coverage.
4. Performance may depend on the underlying communication links.

Difference Between LAN, MAN and WAN

Feature	LAN	MAN	WAN
Full Form	Local Area Network	Metropolitan Area Network	Wide Area Network
Coverage	Small area	City or metropolitan area	Very large geographical area
Example	College lab	City-wide organization network	Internet
Speed	Generally high	Generally high to moderate	Varies
Ownership	Usually private	Private or service provider	Multiple organizations/service providers
Cost	Low	Medium	High
Complexity	Low	Medium	High

COMPLETE UNIT REVISION

Important Definitions

Computer Network

A collection of interconnected devices that communicate and share resources.

Signal

An electrical or electromagnetic representation used to carry information.

Analog Signal

A continuous signal that changes smoothly over time.

Digital Signal

A discrete signal that represents data using distinct values, commonly 0 and 1.

Parallel Transmission

Transmission in which multiple bits are sent simultaneously using multiple lines.

Serial Transmission

Transmission in which bits are sent one after another over a communication channel.

Transmission Media

The path through which data travels from sender to receiver.

Guided Media

Transmission media that uses a physical path such as cable.

Unguided Media

Transmission media that uses wireless signals through air or space.

Network Topology

The physical or logical arrangement of devices and communication links in a network.

LAN

A network covering a small geographical area.

MAN

A network covering a city or metropolitan area.

WAN

A network covering a large geographical area.

QUICK REVISION TABLE

Topic	Key Point
Computer Network	Connects devices for communication and resource sharing
Analog Signal	Continuous signal
Digital Signal	Discrete signal, commonly 0 and 1
Parallel Transmission	Multiple bits transmitted simultaneously
Serial Transmission	Bits transmitted one after another
Twisted Pair	Copper cable with twisted wires
Coaxial Cable	Shielded copper-based cable
Optical Fiber	Uses light for data transmission
Radio Waves	Wireless electromagnetic communication
Bus Topology	Single shared backbone
Star Topology	Central switch/hub
Ring Topology	Circular connection
Mesh Topology	Multiple interconnections
LAN	Small geographical area
MAN	City/metropolitan area
WAN	Large geographical area

IMPORTANT EXAM QUESTIONS

Long Answer Questions

1. What is a computer network? Explain its components and characteristics.
2. Explain the advantages and disadvantages of computer networking.
3. What is a signal? Explain analog and digital signals in detail.
4. Differentiate between analog and digital signals.
5. Explain the characteristics of analog and digital signals.
6. Explain parallel and serial transmission modes with examples.
7. Differentiate between parallel and serial transmission.
8. What is transmission media? Explain guided and unguided transmission media.
9. Explain twisted-pair, coaxial, and optical fiber cables.

10. Explain radio waves, microwaves, and infrared transmission.
 11. Differentiate between guided and unguided transmission media.
 12. What is network topology? Explain Bus topology with advantages and disadvantages.
 13. Explain Star topology with its advantages and disadvantages.
 14. Explain Ring topology with its advantages and disadvantages.
 15. Explain Mesh topology with its advantages and disadvantages.
 16. Compare Bus, Star, Ring, and Mesh topologies.
 17. What is LAN? Explain its characteristics and advantages.
 18. What is MAN? Explain its characteristics.
 19. What is WAN? Explain its characteristics, advantages, and disadvantages.
 20. Differentiate between LAN, MAN, and WAN.
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SHORT ANSWER QUESTIONS

1. Define computer networking.
2. What is a signal?
3. Define analog signal.
4. Define digital signal.
5. What is amplitude?
6. What is frequency?
7. What is phase?
8. What is parallel transmission?
9. What is serial transmission?
10. What is transmission media?
11. Name two guided transmission media.
12. Name two unguided transmission media.
13. What is twisted-pair cable?
14. What is coaxial cable?
15. What is optical fiber?
16. What is network topology?
17. What is Bus topology?
18. What is Star topology?

19. What is Ring topology?
 20. What is Mesh topology?
 21. Define LAN.
 22. Define MAN.
 23. Define WAN.
 24. Give one example of a LAN.
 25. Give one example of a WAN.
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VERY IMPORTANT DIFFERENCES FOR EXAM

- 1. Analog Signal vs Digital Signal**
 - 2. Parallel Transmission vs Serial Transmission**
 - 3. Guided Media vs Unguided Media**
 - 4. Bus Topology vs Star Topology**
 - 5. Star Topology vs Ring Topology**
 - 6. LAN vs MAN**
 - 7. MAN vs WAN**
 - 8. LAN vs WAN**
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CONCLUSION

Computer networks are essential for modern communication and resource sharing. Data is transmitted using signals through various transmission media. Networks can use different topologies depending on their requirements, and they can be classified into LAN, MAN, and WAN based on geographical coverage.

Understanding **signals, transmission modes, transmission media, network topologies, and network types** provides the foundation for studying advanced computer networking concepts.